Daikin BOM Compare — Presales Demo Build Plan

Project: Profisee MDM – BOM Diff / Compare Custom Portal POC
Client: Daikin Comfort Technologies North America (Comfort Tech)
Owner: Stephen Vick (Profisee presales)
Plan Date: February 18, 2026

Executive Summary

This plan details the build of a **Daikin-branded, single-page React/TypeScript web app** that demonstrates side-by-side BOM version comparison with delta highlighting, designed to be embedded as an iFrame alongside Profisee's MDM portal. The app will use mock data modeled on realistic HVAC product structures (chillers, air handling units) and be architected so the mock API layer can be swapped for Profisee REST endpoints in production.

The deliverable is a **presales demo asset** — it must look production-grade, feel fast with large datasets, and tell Daikin's story back to them: configure-to-order BOMs flowing from PLM → MDM → ERP, with MDM as the governance and visibility layer.

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1. Brand Identity & Design System

1.1 Color Tokens (from Daikin Visual Identity)

Token	Hex	Usage
`--daikin-blue`	#00A0E4	Primary actions, active states, links, selected tabs
`--daikin-arctic`	#44C8F5	Hover states, secondary accent, progress indicators
`--daikin-black`	#231F20	Primary text, header bar background, tree connectors
`--daikin-white`	#FFFFFF	Page background, card surfaces
`--daikin-light-blue-grey`	#F1F4F7	Panel backgrounds, alternating rows
`--daikin-caring-blue`	#BEDCF4	Info badges, light highlights
`--daikin-caring-grey`	#ECEAEA	Borders, dividers, disabled states
`--diff-added`	#16A34A (green-600)	Added node badge + left-border accent

Token	Hex	Usage
`--diff-removed`	#DC2626 (red-600)	Removed node badge + strikethrough text
`--diff-changed`	#F59E0B (amber-500)	Changed node badge + attribute highlight
`--diff-moved`	#8B5CF6 (violet-500)	Moved node badge (stretch goal)

1.2 Typography

Role	Font	Weight	Size
App header / logo text	**Barlow Semi Condensed** (Google Fonts)	700	18px
Section headings	Barlow Semi Condensed	600	15px
Body / tree node labels	**Barlow**	400/500	13–14px
Monospace (part numbers, qty)	**JetBrains Mono**	400	12px
Metadata captions	Barlow	400	11px

Rationale: Barlow's condensed industrial feel maps well to Daikin's engineering/manufacturing audience. It's distinctive without being decorative. JetBrains Mono for part numbers adds technical credibility.

1.3 Visual Identity Elements

- **Trapezoid accent shape** — Daikin's brand uses a rectangular trapezoid (one slanted side, orientation up-right) as a graphic motif. We'll use a subtle CSS clip-path trapezoid on the header bar's right edge and as a decorative element on empty states.
- **Logo placement** — Daikin triangle mark + "DAIKIN" logotype, top-left of header bar, on dark (`--daikin-black`) background. White logo variant.
- **No rounded corners > 4px** — keep the UI sharp and industrial. 2px border-radius on inputs/buttons, 4px on cards.

1.4 Component Style Guide

Component	Spec
Buttons (primary)	`--daikin-blue` fill, white text, 2px radius, 36px height, Barlow 600
Buttons (secondary)	White fill, `--daikin-blue` border + text

Component	Spec
Select dropdowns	36px height, light grey border, blue focus ring
Badges	Pill shape (12px radius), 10px font, colored per diff type
Tree node row	32px height, 12px left-indent per level, hover = `--daikin-light-blue-grey`
Cards	White surface, 1px `--daikin-caring-grey` border, 4px radius, 16px padding
Tabs	Underline style, `--daikin-blue` active indicator (3px), no background fill

2. Information Architecture & Screen Layout

2.1 App Shell

```
┌──────────────────────────────────────────────────────────────────────────┐
│ [▲ DAIKIN]   BOM Compare Portal        [? Help] [User] │ ← Header bar (48px, --daikin-black)
├──────────────────────────────────────────────────────────────────────────┤
│ Product: [DX20VC-060 ▼]  BOM A: [E-BOM v3.2 ▼]        │ ← Selector bar (56px)
│                    BOM B: [M-BOM v1.0 ▼] [Compare]│
├──────────────────────────────────────────────────────────────────────────┤
│ [Side-by-Side]  [Delta Summary]  [Metadata]            │ ← Tab bar
├──────────────────────────────────────────────────────────────────────────┤
│                                        │
│  (Active tab content — see below)                    │ ← Main workspace
│                                        │
├──────────────────────────────────────────────────────────────────────────┤
│ 18 differences across 2,413 nodes  •  Last compared: now │ ← Status footer (28px)
└──────────────────────────────────────────────────────────────────────────┘
```

2.2 Tab 1: Side-by-Side View

```
┌──────────────────────────────────────┬──┬──────────────────────────────────────┐
│ BOM A: E-BOM v3.2        │ │ BOM B: M-BOM v1.0        │
│ [Search 🔍] [± Expand]   │ │ [Search 🔍] [± Expand]   │
├──────────────────────────────────────┤ ├──────────────────────────────────────┤
│ ▼ DX20VC-060             │ │ ▼ DX20VC-060             │
│   ▼ Compressor Assy      │ │   ▼ Compressor Assy      │
│     ● ZP51K5E  (1)       │ │     ● ZP54K5E  (1) [CHG]│
│     ● 326-123  (2)       │ │     ● 326-123  (2)       │
│   ▶ Condenser Coil       │ │   ▶ Condenser Coil       │
```

```
│ ▼ Control Board      │ │ ▼ Control Board      │
│   ● PCB-A100  (1)    │ │   ● PCB-A100  (1)    │
│   ● CAP-220   (3)    │ │   ○ (removed)   [DEL]│
│                      │ │   ● CAP-330   (2) [ADD]│
│ ...                  │ │ ...                  │
└──────────────────────┘ └──────────────────────┘

↕ Synchronized scrolling (toggle)
```

Key interactions:

- Trees scroll independently by default; toggle to sync scroll
- Clicking a node in one tree highlights the matched node (or gap) in the other
- "Show differences only" toggle collapses unchanged subtrees
- Breadcrumb path shown above trees for current expanded context
- Diff badges (Added / Removed / Changed) appear as colored pills on affected nodes
- Changed attributes show inline: Qty: 3 → 2, Part: ZP51K5E → ZP54K5E

2.3 Tab 2: Delta Summary

```
┌──────────────────────────────────────────────────────────────────────┐
│ Filter: [All ▼]  [Added: 4] [Removed: 2] [Changed: 12]│
├──────┬──────────┬──────────┬──────────┬──────────────────────┤
│ Type│ Path     │ Attribute│ BOM A    │ BOM B                │
├──────┼──────────┼──────────┼──────────┼──────────────────────┤
│ CHG │ /Comp/.. │ Part No  │ ZP51K5E  │ ZP54K5E              │
│ CHG │ /Comp/.. │ Qty      │ 3        │ 2                    │
│ DEL │ /Ctrl/.. │ Component│ CAP-220  │ —                    │
│ ADD │ /Ctrl/.. │ Component│ —        │ CAP-330              │
│ ... │          │          │          │                      │
└──────┴──────────┴──────────┴──────────┴──────────────────────┘

Click row → navigates to node in Side-by-Side view
```

2.4 Tab 3: Metadata Panel

Two cards side-by-side showing BOM header metadata:

Field	BOM A	BOM B
Version	E-BOM v3.2	M-BOM v1.0
Source System	Windchill PLM	SAP S/4HANA
Status	Released	In Review
Effective Date	2026-01-15	2026-02-10
Owner	J. Tanaka (Engineering)	M. Rodriguez (Manufacturing)
Total Components	2,413	2,398
Change Summary	—	18 differences from E-BOM v3.2

Plus a "View in Profisee →" deep-link button on each card (mock, demonstrates the integration narrative).

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3. Mock Data Architecture

3.1 Product Catalog (2 products)

Product ID	Model	Description	Use Case
PROD-001	DX20VC-060	5-Ton Variable-Speed Split System Heat Pump	Large residential/light commercial — moderate BOM (~150 nodes)
PROD-002	AWS-CHL-500T	500-Ton Centrifugal Water-Cooled Chiller	Custom-engineered commercial chiller — **large BOM (~2,500 nodes)** for virtualization demo

3.2 BOM Versions Per Product

Each product gets **3 BOM versions**:

BOM Type	Source System	Editable in MDM?	Notes
E-BOM (Engineering)	Windchill PLM	No (view only)	Original engineering release
M-BOM (Manufacturing)	SAP S/4HANA	No (view only)	Manufacturing-optimized (phantom assemblies resolved, tooling added)
Service BOM	Profisee MDM	**Yes**	Field-maintainable version with replacement parts, service kits

3.3 BOM Node Schema

```typescript
```

```typescript
interface BOMNode {
  id: string;                    // Stable line ID (e.g., "LN-00412")
  parentId: string | null;       // Parent node ID (null = root)
  level: number;                 // Nesting depth (0 = root)
  position: number;              // Sort order within parent
  componentId: string;           // Part number (e.g., "ZP54K5E-PFV")
  componentName: string;         // Description
  quantity: number;
  uom: string;                   // Unit of measure
  referenceDesignator?: string;  // Position reference on assembly
  lifecycleStatus: 'Active' | 'Obsolete' | 'Pending' | 'Prototype';
  isReplaceable: boolean;        // Can be field-replaced
  category: 'Assembly' | 'Subassembly' | 'Component' | 'Raw Material' | 'Consumable';
  supplier?: string;
  leadTimeDays?: number;
  warrantyMonths?: number;
  lastModified: string;          // ISO date
  modifiedBy: string;
}

interface BOMVersion {
  bomId: string;
  productId: string;
  bomType: 'E-BOM' | 'M-BOM' | 'Service';
  version: string;
  sourceSystem: string;
  status: 'Draft' | 'In Review' | 'Released' | 'Superseded';
  effectiveDate: string;
  owner: string;
  ownerRole: string;
  totalNodes: number;
  nodes: BOMNode[];
}
```

3.4 HVAC-Realistic Component Taxonomy

The mock data should use real-feeling HVAC component categories. For the chiller (PROD-002):

```
AWS-CHL-500T (Root Assembly)
├── Compressor Section
│   ├── Centrifugal Compressor (York/JCI style part numbers)
│   ├── Variable Speed Drive (VSD)
│   ├── Motor Assembly
│   ├── Bearings & Seals
│   └── Lubrication System
├── Evaporator Section
│   ├── Shell & Tube Heat Exchanger
│   ├── Refrigerant Distribution
│   └── Tube Bundle
├── Condenser Section
│   ├── Shell & Tube Condenser
│   ├── Water Box Assemblies
│   └── Relief Valves
├── Controls & Electrical
│   ├── Main Control Panel (MCP)
│   ├── Starter Assembly
│   ├── Sensors & Transducers
│   └── Wiring Harnesses
├── Refrigerant Circuit
│   ├── Expansion Devices
│   ├── Piping & Fittings
│   ├── Filter Driers
│   └── Sight Glasses
├── Structural / Frame
│   ├── Base Frame
│   ├── Isolation Mounts
│   └── Acoustic Enclosure Panels
└── Documentation & Kits
    ├── Service Manual Kit
    ├── Commissioning Kit
    └── Spare Parts Kit
```

3.5 Diff Scenarios to Seed

For the demo to tell a compelling story, the mock data includes these pre-seeded differences:

Scenario	BOM A	BOM B	Diff Type	Story
Compressor upgrade	ZP51K5E	ZP54K5E (higher capacity)	Changed	Engineering spec'd one; manufacturing upgraded for availability
Capacitor substitution	CAP-220	CAP-330	Changed	Component substitution during manufacturing
Added tooling fixture	—	JIG-CHK-001	Added	Manufacturing BOM includes tooling not in E-BOM
Removed prototype sensor	SENS-PROTO-X1	—	Removed	Prototype sensor removed before manufacturing release
Quantity change	Bolt M8x40 qty=24	Bolt M8x40 qty=28	Changed	Manufacturing found additional fastener points
Service kit added	—	SVC-KIT-500T-A	Added	Service BOM adds field service kits
Warranty extension part	Bearing (12mo)	Bearing (24mo warranty)	Changed	Service BOM extends warranty on wear item

4. Technical Architecture

4.1 Stack

Layer	Technology	Rationale
Framework	**React 18** + TypeScript	Matches handoff spec; component-model fits tree virtualization
Build	**Vite**	Fast dev iteration for demo
Styling	**Tailwind CSS** + CSS custom properties	Rapid layout with Daikin brand tokens
Tree virtualization	**react-window** or custom virtual list	Render only visible nodes from 2,500+ node BOMs
Diff engine	**Custom (in-memory)**	Precompute diff map on BOM pair selection

Layer	Technology	Rationale
State management	**React Context + useReducer**	Lightweight; compare state serializable to URL params
Fonts	Google Fonts (Barlow, JetBrains Mono)	CDN loaded
Icons	**Lucide React**	Clean, consistent icon set

4.2 Module Structure

```
src/
├── api/
│   ├── types.ts              # BOMNode, BOMVersion, Product interfaces
│   ├── BOMService.ts         # Abstract service interface
│   ├── MockBOMService.ts     # Mock implementation with static JSON
│   └── ProfiseeBOMService.ts # Stub for future Profisee REST adapter
├── data/
│   ├── products.json
│   ├── bom-dx20vc-ebom.json
│   ├── bom-dx20vc-mbom.json
│   ├── bom-dx20vc-service.json
│   ├── bom-chl500t-ebom.json # Large BOM (~2,500 nodes)
│   ├── bom-chl500t-mbom.json
│   └── bom-chl500t-service.json
├── diff/
│   ├── computeDiff.ts        # Core diff algorithm
│   ├── diffTypes.ts          # DiffResult, DiffEntry types
│   └── nodeIdentity.ts       # Stable node matching logic
├── components/
│   ├── layout/
│   │   ├── AppHeader.tsx
│   │   ├── SelectorBar.tsx
│   │   ├── TabBar.tsx
│   │   └── StatusFooter.tsx
│   ├── tree/
│   │   ├── BOMTree.tsx       # Virtualized tree with diff overlay
│   │   ├── BOMTreeNode.tsx   # Single row renderer
│   │   ├── TreeConnectors.tsx # Visual indent lines
│   │   └── DiffBadge.tsx
│   ├── compare/
│   │   ├── SideBySideView.tsx # Two-panel tree compare
│   │   ├── DeltaSummary.tsx   # Table of differences
```

```
│   │   └── MetadataPanel.tsx    # BOM header comparison cards
│   ├── shared/
│   │   ├── SearchInput.tsx
│   │   ├── FilterChips.tsx
│   │   └── DeepLinkButton.tsx   # "View in Profisee →"
│   └── brand/
│       ├── DaikinLogo.tsx       # SVG logo component
│       └── TrapezoidAccent.tsx  # Brand shape element
├── hooks/
│   ├── useCompareState.ts       # Product/BOM selection + diff result
│   ├── useVirtualTree.ts        # Virtualization logic
│   └── useSyncScroll.ts         # Synchronized scroll between panels
├── context/
│   └── CompareContext.tsx       # Global compare state
├── theme/
│   └── daikin-tokens.css        # CSS custom properties
├── App.tsx
└── main.tsx
```

4.3 Diff Algorithm Design

Node identity matching: Nodes are matched between BOM A and BOM B using a composite key:

```
identity = parentPath + "/" + componentId + "@" + position
```

If a stable line ID (`id`) is present and shared across BOM versions, prefer that. Fallback to the composite key.

Diff computation (precomputed on compare action):

1. Flatten both trees into maps keyed by identity
2. For each node in BOM A: if missing from BOM B → `Removed`
3. For each node in BOM B: if missing from BOM A → `Added`
4. For matched nodes: compare `quantity`, `componentId`, `componentName`, `lifecycleStatus`, `supplier`, `warrantyMonths` → if any differ → `Changed` with per-attribute detail
5. Output a `DiffMap`: `Map<nodeIdentity, DiffEntry>`

```typescript
```

```typescript
interface DiffEntry {
  type: 'added' | 'removed' | 'changed' | 'unchanged';
  nodeA?: BOMNode;
  nodeB?: BOMNode;
  changes?: AttributeChange[];  // For 'changed' type
}

interface AttributeChange {
  field: string;
  valueA: string | number;
  valueB: string | number;
}
```

4.4 Tree Virtualization Strategy

For the 2,500-node chiller BOM:

- Maintain a **flat expanded-node list** (only nodes whose ancestors are all expanded)
- Render with a **virtual window** — only mount DOM nodes within viewport + 20-node overscan
- Each row is a fixed 32px height → total scroll height is deterministic
- Expand/collapse updates the flat list; virtual window re-renders only visible range
- Target: **< 16ms per frame** during scroll on modern hardware

4.5 Service Abstraction (Integration-Ready)

```typescript
interface BOMService {
  getProducts(): Promise<Product[]>;
  getBOMVersions(productId: string): Promise<BOMVersionSummary[]>;
  getBOM(bomId: string): Promise<BOMVersion>;
}

// POC uses MockBOMService; production swaps to:
// ProfiseeBOMService (calls Profisee REST: GET /api/v1/members/{entityId}/...)
```

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5. Build Phases & Execution Order

Phase 1: Foundation (estimated: ~2 hours)

1. **Scaffold React + Vite + TypeScript project**
2. **Create** `daikin-tokens.css` with all brand colors, typography, spacing tokens
3. **Build** `AppHeader` with Daikin logo (SVG), title, trapezoid accent
4. **Build** `SelectorBar` — Product dropdown, BOM A dropdown, BOM B dropdown, Compare button
5. **Build** `TabBar` and `StatusFooter` shell
6. **Define TypeScript interfaces** (`types.ts`, `diffTypes.ts`)

Phase 2: Mock Data (estimated: ~1 hour)

7. **Create product catalog JSON** (2 products)
8. **Create 6 BOM JSON files** — 3 per product, with HVAC-realistic part numbers and hierarchy
9. **Build large chiller BOM generator** — script to generate ~2,500 nodes with realistic nesting
10. **Seed diff scenarios** (the 7 pre-planned differences per BOM pair)
11. **Implement** `MockBOMService` with JSON loading

Phase 3: Diff Engine (estimated: ~1 hour)

12. **Implement** `nodeIdentity.ts` — composite key generation
13. **Implement** `computeDiff.ts` — the full diff algorithm
14. **Write** `useCompareState` **hook** — wires selector changes → diff computation → state

Phase 4: Side-by-Side View (estimated: ~2 hours)

15. **Build** `BOMTree` with virtualization (flat list approach)
16. **Build** `BOMTreeNode` — indented row with expand/collapse, part number, qty, diff badge
17. **Build** `TreeConnectors` — visual indent lines connecting parent-child
18. **Build** `DiffBadge` — colored pill (Added/Removed/Changed)
19. **Implement** `SideBySideView` — two `BOMTree` panels with resizable divider
20. **Add synchronized scrolling** (`useSyncScroll`)
21. **Add "Show differences only" toggle** — collapses unchanged subtrees
22. **Add search-within-tree** functionality

Phase 5: Delta Summary (estimated: ~1 hour)

23. **Build** `DeltaSummary` table with columns: Type, Path, Attribute, BOM A Value, BOM B Value
24. **Add filter chips** (All / Added / Removed / Changed) with counts
25. **Implement click-to-navigate** — clicking a delta row switches to Side-by-Side and scrolls to that node

Phase 6: Metadata & Polish (estimated: ~1 hour)

26. **Build** `MetadataPanel` — two side-by-side cards with BOM header details
27. **Add "View in Profisee →" deep link buttons** (mock URL)
28. **Add change density indicator** in status footer ("18 differences across 2,413 nodes")
29. **Add empty state** (before compare is triggered) — Daikin trapezoid motif with instructional text
30. **Responsive check** — ensure iFrame-friendly sizing (no hardcoded full-screen)

Phase 7: Demo Script & Packaging (estimated: ~30 min)

31. **Write demo script** (2–3 minute walkthrough, see Section 7)
32. **Build production bundle** (`vite build`)
33. **Test in iFrame container** (simple wrapper HTML page simulating Profisee portal embed)

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6. Key Presales Demo Moments

The demo should hit these beats in roughly this order (2–3 minutes):

Beat 1: "Single stop for all BOM versions" (30 sec)

- Open the app → select product **DX20VC-060**
- Show that all three BOM types (E-BOM, M-BOM, Service) are available from dropdowns
- Talking point: *"Daikin loads BOMs from PLM and ERP into Profisee as the central visibility layer. Data stewards see everything here — no jumping to Windchill or S/4."*

Beat 2: "Side-by-side compare at scale" (45 sec)

- Select E-BOM vs M-BOM → click Compare
- Trees render side-by-side with diff badges immediately visible
- Drill into Compressor Section → show the compressor part number change highlighted
- Talking point: *"The compare UI is custom because Daikin needs split-screen + delta at scale — but it's still calling Profisee APIs behind the scenes, using the same security model, and can deep-link back into*

Profisee for DQ rules, workflows, and audit."

Beat 3: "Delta summary for quick triage" (30 sec)

- Switch to Delta tab → show filtered list of all 18 differences
- Click "Changed" filter chip → narrows to only attribute changes
- Click a row → auto-navigates to Side-by-Side view, scrolled to that node
- Talking point: *"When an engineering release happens in PLM, Profisee applies DQ checks and approvals, then triggers downstream publishing to S/4. This delta view lets stewards quickly triage what changed."*

Beat 4: "Scales to real-world complexity" (30 sec)

- Switch to product **AWS-CHL-500T** (the chiller)
- Compare E-BOM vs Service BOM → 2,500 nodes render instantly (virtualization)
- Show the "18 differences across 2,413 nodes" density indicator
- Fast scroll through tree — stays smooth
- Talking point: *"This is a 2,500-line chiller BOM. The UI virtualizes rendering so only visible nodes hit the DOM — no performance cliff as BOMs grow."*

Beat 5: "Deep link back to Profisee" (15 sec)

- Click "View in Profisee →" button on metadata card
- (Mock behavior — show a toast/modal saying "Opens Profisee record view for ZP54K5E")
- Talking point: *"One click takes you to the full Profisee record for any component — DQ history, workflow state, cross-references, the full MDM picture."*

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7. Risks & Mitigations

Risk	Mitigation
Mock data doesn't feel "real" to Daikin engineers	Use HVAC-standard terminology, realistic part number formats (e.g., Copeland ZP series compressors, standard condenser coil assemblies). Cross-reference with Daikin product catalogs.
Virtualized tree has visual glitches	Fallback: use a simpler paginated tree (load 200 nodes at a time) if full virtualization proves brittle in iFrame context.

Risk	Mitigation
Brand colors feel too plain for a demo	Add subtle depth: light gradient on header, shadow on floating panels, microanimation on diff badge appearance (fade+scale in).
Daikin asks "is this real data?"	Stephen should pre-frame: "This is a representative mock to show the UX approach and integration architecture. Real data would connect to your Windchill/S4/Profisee instances."
Over-selling OOTB	The plan explicitly positions this as a custom UI layer on top of Profisee's API/data/governance platform. Never suggest Profisee portal does side-by-side natively.

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8. Open Items Carried from Handoff

These are Daikin's explicit asks from the Feb 17 meeting — tracked here for Stephen's awareness:

#	Item	Status	Owner
1	Customer references (clients with custom Profisee portals)	Pending — Nick to source 1–2 references	Nick / JW
2	Mock-up showing side-by-side BOM + "behind the scenes" explanation	**This POC**	Stephen / Dev team
3	Directional SOW (day 1–90) with cost for custom UI component	Pending	John / Stephen
4	Follow-up meeting next week	Pending — send time slots once mockup + refs ready	Nick

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9. Success Criteria

The demo is successful if Daikin walks away confident that:

1. **Multiple BOM versions can coexist in Profisee** and be visualized from a single interface
2. **The custom portal is a thin UI layer**, not a parallel system — data, governance, security, and integration remain in Profisee
3. **The approach scales** to their configure-to-order reality (thousands of nodes, frequent iteration)
4. **It looks and feels like their tool** — Daikin brand, HVAC terminology, their workflow language

5. **Profisee upgrades won't break it** — because the portal only consumes stable REST APIs

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Ready to build. Next step: scaffold the project and begin Phase 1.